Recursive Think-Answer Process for LLMs and VLMs

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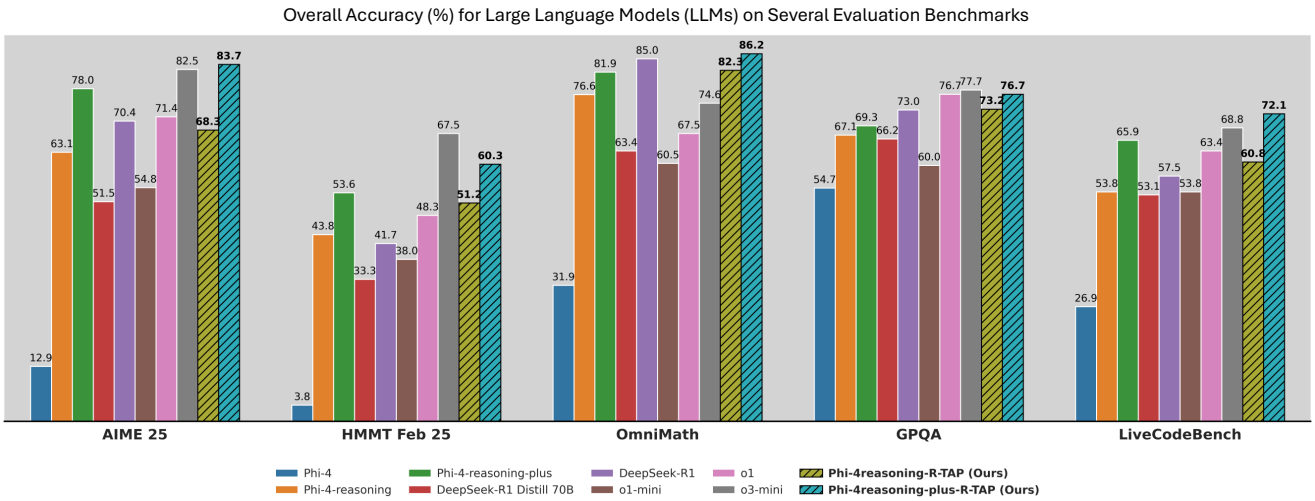

Figure 1. Overall accuracy (%) of numerous large language models (LLMs) on five evaluation benchmarks—AIME25 [65], HMMT Feb 25 [2], OmniMath [29], GPQA [25], and LiveCodeBench [43].

Abstract

Think–Answer reasoners such as DeepSeek-R1 have made notable progress by leveraging interpretable internal reasoning. However, despite the frequent presence of self-reflective cues like "Oops!", they remain vulnerable to output errors during single-pass inference. To address this limitation, we propose an efficient Recursive Think–Answer Process (R-TAP) that enables models to engage in iterative reasoning cycles and generate more accurate answers, going beyond conventional single-pass approaches. Central to this approach is a confidence generator that evaluates the certainty of model responses and guides subsequent improvements. By incorporating two complementary rewards—Recursively Confidence Increase Reward and Final Answer Confidence Reward—we show that R-TAP-enhanced models consistently outperform conventional single-pass methods for both large language models (LLMs) and vision-language models (VLMs). Moreover, by analyzing the frequency of "Oops"-like expressions in model responses, we find that R-TAP–applied models exhibit significantly fewer self-reflective patterns, resulting in more stable and faster inference-time reasoning. We hope R-TAP pave the way evolving into efficient and elaborated methods to refine the reasoning processes of future AI.

1. Introduction

Recent advances in a Think–Answer process-equipped models—such as OpenAI's o1 [42] and DeepSeek-R1 [35]—have demonstrated that explicitly separating the reasoning and answering stages can substantially enhance problem-solving performance. By following a Think–Answer scheme, these models surpass direct-prediction approaches and achieve remarkable capabilities in domains such as mathematical reasoning and competitive programming. This paradigm has also been extended to multimodal settings, where recent VLMs leverage a Think-

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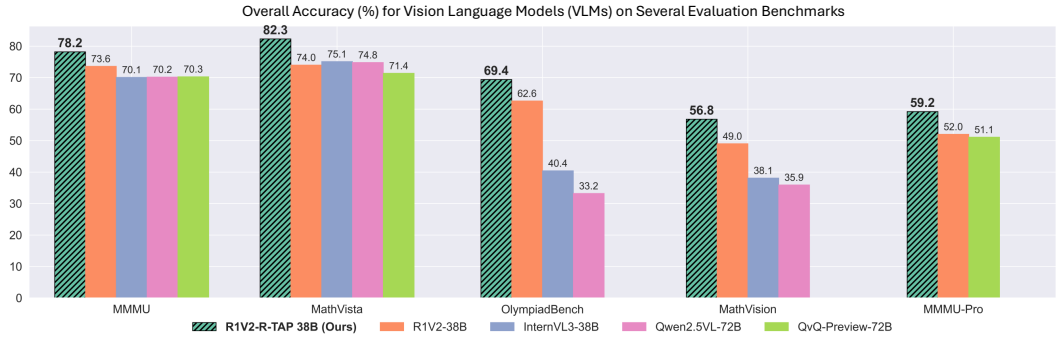


Figure 2. Overall accuracy (%) of numerous vision language models (VLMs) on five evaluation benchmarks—MMMU [94], MathVista [58], OlympiadBench [36], MathVision [78], and MMMU-Pro [95].

SYSTEM: You are a helpful AI Assistant that provides well-reasoned and detailed responses. You first think about the reasoning process as an internal monologue and then provide the user with the answer. Respond in the following format: <think> </think>\n<answer> </answer>

USER: Albert writes 2025 numbers a_{1}, \ldots, a_{2025} in a circle on a blackboard. Initially, each of the numbers is uniformly and independently sampled at random from the interval [0,1]. Then, each second, he \emph{simultaneously} replaces a_{i} with \max \left(a_{i-1}, a_{i}, a_{i+1}\right) for all i=1,2, \ldots, 2025 (where a_{0}=a_{2025} and a_{2026}=a_{1}). Compute the expected value of the number of distinct values remaining after 100 seconds.

ASSISTANT: <think> We want the expected number of distinct values remaining after 100 synchronous "max-with-neighbors" updates on a circle of length 2025. After \(t\) steps, each final value is the maximum over some interval of the initial samples. A natural first guess is that the "influence window" of each initial point expands one step per second in both directions, giving a window length of \(2t+1\). If these windows partition the circle roughly evenly, one might expect about \[\frac{2025}{2\cdot100 + 1} ~=~ \frac{2025}{201} \] distinct values, since each block of length 201 contributes one surviving maximum on average. </think>
<answer> \frac{2025}{201} </answer>

<think> **Ah—on reflection**, I realize that because we update simultaneously, the propagation of a crest might not spread symmetrically in both directions in exactly the same way I assumed. Instead, each step seems to effectively pull maxima inwards by one, but maybe only one-sided propagation of length 100 matters, giving window length 100 rather than 201. If the relevant block size is really \(t\) (not \(2t+1\)), then we'd predict \[\frac{2025}{100} \] distinct values remaining. </think>
<answer> \frac{2025}{100} </answer>

<think> **I made another off-by-one error.** In fact, each initial peak influences its neighbors one step per second on each side, but since we include the center itself at time 0, the true window length after \(t\) seconds is \(t+1\) in each direction—yet because it's circular, one only counts distinct windows of length \(t+1\) before they overlap. The well-known result for this max–smoothing process on a circle is that one ends up with about \(n/(t+1)\) survivors. Here \(n=2025\) and \(t=100\), so the correct expected number is \[\frac{2025}{100+1} = \frac{2025}{101}. \] </think>
<answer> \frac{2025}{101} </answer>

Figure 3. Qualitative example of recursive think–answer process on a combinatorics question. The model iteratively refines its solution across multiple reasoning cycles, successfully correcting initial misconceptions such as off-by-one errors.

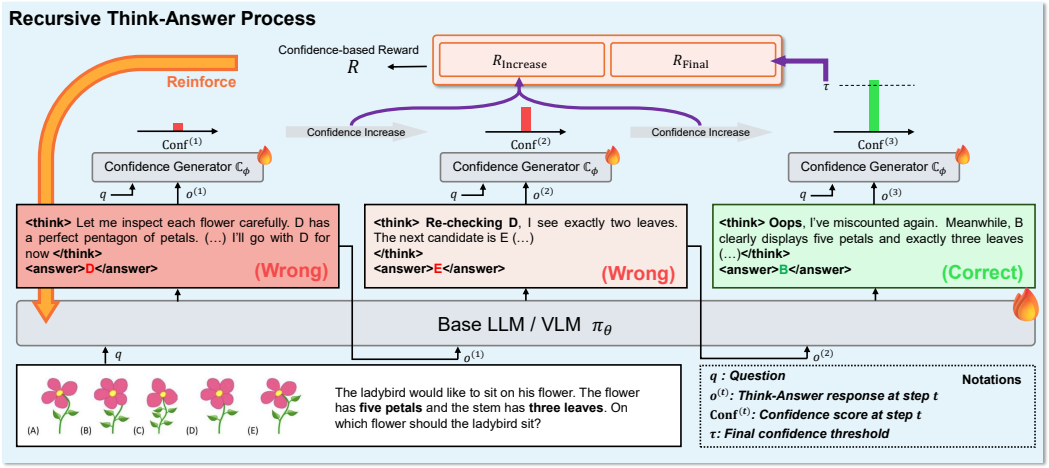
Answer process [16, 24, 40, 62, 66, 89, 102] to improve vision-language reasoning.

Despite these successes, current Think–Answer models almost always rely on a single-pass reasoning trajectory. After generating one Think–Answer pair, the model stops its inference process—even when the reasoning is inaccurate, inconsistent, or clearly uncertain. Models often produce self-reflective cues such as "Oops!" or "Let me try again" which show their uncertainty. However, these signals are not used: the model outputs its final answer without any method for self-evaluation or additional refinement. As a result, incorrect but confident-sounding reasoning remains uncorrected, reducing the reliability and consistency of Think–Answer models.

This limitation mainly comes from current reinforcement learning (RL) frameworks. Recent GRPO-style [70] sampling methods optimize only a single Think–Answer trajectory with rewards such as accuracy or format correctness. However, these methods do not consider the model's confidence in its reasoning, so they cannot support introspective checks or recursive correction. Because of this, the model cannot tell when its answer has low confidence and needs further refinement, which becomes a serious issue for complex or high-stakes tasks.

To address these limitations, we introduce R-TAP, **R**ecursive **T**hink-**A**nswer **P**rocess that enables LLMs and VLMs to iteratively refine their reasoning through confidence-guided Think–Answer cycles. Instead of stopping after one reasoning pass, the model evaluates its own confidence after each cycle using a dedicated *Confidence Generator*. When the models internally think the confidence is low, the model runs an additional Think–Answer cycle, revisiting and improving its previous reasoning. This recursive approach encourages deeper thinking, self-reflection, and step-by-step refinement of the reasoning process.

R-TAP introduces two key technical components. First is a Confidence Generator, initialized from the pretrained model and fine-tuned to quantify correctness of individual reasoning paths. Second is a recursive reward structure, combining (i) *Recursively Confidence Increase Reward* that encourages confidence to improve from one cycle to the next, and (ii) *Final Answer Confidence Reward* that encourages high-confidence final predictions. Together,



(Wrong)

Figure 4. **Recursive Think-Answer Process.** Given a question q, base LLM/VLM π_θ recursively generates multiple Think-Answers $o^{(t)}$ until the answer is correct $t = M$. In this example, effective recursion depth $M = 3$. A pre-trained Confidence Generator $\mathbb{C}_\phi$ assess each question and Think-Answer pair $(q, o^{(t)})$ then generates confidence score $\text{Conf}^{(t)}$. This confidence score is used to formulate confidence-based reward – R_{Increase} and R_{Final} – which serves as a sufficient reinforcement signal to train the model to recursively generate higher confidence Think-Answers until intrinsic confidence is high enough. Note that full responses for this question is described in Appendix A.

these components provide the necessary training signals for LLMs and VLMs to learn recursive Think-Answer trajectories for strengthening their reasoning.

To implement this approach, we proceed in two stages. Stage 1 performs supervised learning on the Confidence Generator using binary correctness labels for each reasoning trajectory produced by the target model. Stage 2 applies RL with GRPO to optimize the model's reasoning behavior under recursive rewards, enabling the model to generate progressively more accurate and confident reasoning across cycles. During this stage, the Confidence Generator is trained simultaneously to predict reliable confidence scores for the updated model's responses in real time. Notably, the Confidence Generator is used only during training, so R-TAP introduces no additional inference-time cost. In summary, R-TAP explicitly optimizes both introspection and self-correction, allowing models to refine their own trajectories and move beyond the rigid, one-shot Think-Answer paradigm.

Our experiments show that R-TAP delivers strong and consistent performance improvements across diverse language and vision-language reasoning benchmarks. Moreover, we find that recursive refinement leads to a substantial reduction in "Oops!"-style self-corrections during inference, indicating that R-TAP achieves more reliable yet fast inference-time reasoning with fewer failures along the trajectory. These results demonstrate that confidence-guided recursion training is a powerful mechanism for enhancing both the accuracy and inference speed of modern reasoning models. Our main contributions are summarized as follows:

- **Recursive Think-Answer Process**: We propose R-TAP, a confidence-driven iterative reasoning framework that enables dynamic re-engagement of reasoning cycles and self-corrective refinement.
- **Unified Reasoning Across Modalities**: R-TAP generalizes effectively to both language-only and multimodal reasoning, providing a unified mechanism for recursive improvement.

2. Related Work

Evolution of LLMs and VLMs. The rapid scaling of LLMs and VLMs has enabled impressive improvements in reasoning, alignment, and multimodal understanding. Early efforts such as GPT-3 [13] demonstrated strong in-context learning capabilities, while subsequent alignment-oriented approaches including InstructGPT [64] and ChatGPT leveraged RL from human feedback (RLHF) to improve reliability and adherence to instructions. More advanced closed-source models such as GPT-4 [6] expanded these capabilities to multimodal settings, showcasing strong performance across diverse problem-solving tasks.

Open-source research has paralleled these advances, driven by architectural innovations and large-scale instruction tuning. LLaMA [77] and its successors introduced lightweight yet performant architectures, while models such as LLaVA-NeXT [51], MM1 [60], Yi-VL [92], and MiniGemini [49] scaled VLM reasoning by increasing training data and integrating powerful vision encoders. Additional efforts incorporate specialized projectors or structured modules [15, 30, 76, 91] to extract hierarchical or task-specific features. Despite these advancements, most models still depend on single-pass predictions at inference time, without iterative introspection or self-correction.

Think-Answer Reasoning. Think–Answer paradigm has emerged as a compelling approach for enhancing reasoning in both LLMs and VLMs. Chain-of-Thought prompting [83] first demonstrated the benefits of eliciting inter-

Table 1. Performance comparison of open-source large language models on challenging math benchmarks: AIME24 [65], AMC [1], MATH500 [3], Minerva [4], and OlympiadBench [36].

Base Model + Method	AIME24	AMC	MATH500	Minerva	OlympiadBench	Avg.
Qwen2.5-Math-1.5B [88]	16.7	43.4	61.8	15.1	28.4	33.1
Qwen2.5-Math-1.5B-Instruct [88]	10.0	48.2	74.2	26.5	40.2	39.8
R1-Distill-Qwen-1.5B @ 3k [35]	2.5	21.7	52.2	16.3	17.3	22.0
R1-Distill-Qwen-1.5B @ 8k [35]	20.0	49.4	77.4	25.0	35.8	41.5
R1-Distill-Qwen-1.5B-R-TAP @ 8k	**27.0**	**55.8**	**83.5**	**31.9**	**42.4**	**48.1**
Oat-Zero-1.5B [55]	20.0	53.0	74.2	25.7	37.6	42.1
Oat-Zero-1.5B-R-TAP	**26.2**	**59.7**	**80.0**	**31.2**	**42.8**	**48.0**
OpenReasoner-Zero-7B @ 3k [37]	13.3	47.0	79.2	31.6	44.0	43.0
OpenReasoner-Zero-7B @ 8k [37]	13.3	54.2	82.4	31.6	47.9	45.9
Qwen2.5-Math-7B [88]	0.2	45.8	69.0	21.3	34.7	38.2
Qwen2.5-Math-7B-Instruct [88]	16.7	53.0	83.6	29.8	42.7	45.1
SimpleRL-Zero-7B [97]	26.7	60.2	78.2	27.6	40.3	46.6
PRIME-Zero-7B [23]	16.7	62.7	83.8	36.0	40.9	48.0
R1-Distill-Qwen-7B @ 3k [35]	10.0	26.2	60.1	23.0	23.1	28.5
R1-Distill-Qwen-7B @ 8k [35]	33.3	68.4	88.1	35.9	47.7	54.7
R1-Distill-Qwen-7B-R-TAP	**39.7**	**75.0**	**92.7**	**42.3**	**53.8**	**60.7**
Oat-Zero-7B [55]	43.3	62.7	80.0	30.1	41.0	51.4
Oat-Zero-7B-R-TAP	**50.5**	**69.5**	**87.2**	**37.2**	**46.8**	**57.7**

Table 2. Comparison of closed and open-source language models with MiMo-RL-7B-R-TAB across general knowledge, mathematical, and code benchmarks. General: GPQA Diamond [69], SuperGPQA [25], DROP [26], MMLU-Pro [82], IF-Eval [103]; Math: MATH500 [3], AIME2024 [65], AIME2025 [65]; Code: LiveCodeBench v5, v6 [43].

Benchmark	GPT-4o 0513 [41]	Claude-3.5-Sonnet-1022 [8]	OpenAI o1-mini [42]	QwQ-32B Preview [74]	R1-Distill-Qwen-14B [35]	R1-Distill-Qwen-7B [35]	MiMo-7B-RL [72]	**MiMo-RL 7B-R-TAP**
General								
GPQA Diamond (Pass@1)	49.9	**65.0**	60.0	54.5	59.1	49.1	54.4	60.7
SuperGPQA (Pass@1)	42.4	**48.2**	45.2	43.6	40.6	28.9	40.5	47.3
DROP (3-shot F1)	83.7	**88.3**	83.9	71.2	85.5	77.0	78.7	84.5
MMLU-Pro (EM)	72.6	78.0	**80.3**	52.0	68.8	53.5	58.6	65.9
IF-Eval (Prompt Strict)	84.3	**86.5**	84.8	40.4	78.3	60.5	61.0	68.0
Mathematics								
MATH500 (Pass@1)	74.6	78.3	90.0	90.6	93.9	92.8	95.8	**97.3**
AIME 2024 (Pass@1)	9.3	16.0	63.6	50.0	69.7	55.5	68.2	**75.8**
AIME 2025 (Pass@1)	11.6	7.4	50.7	32.4	48.2	38.8	55.4	**61.9**
Code								
LiveCodeBench v5 (Pass@1)	32.9	38.9	53.8	41.9	53.1	37.6	57.8	**64.2**
LiveCodeBench v6 (Pass@1)	30.9	37.2	46.8	39.1	31.9	23.9	49.3	**56.3**

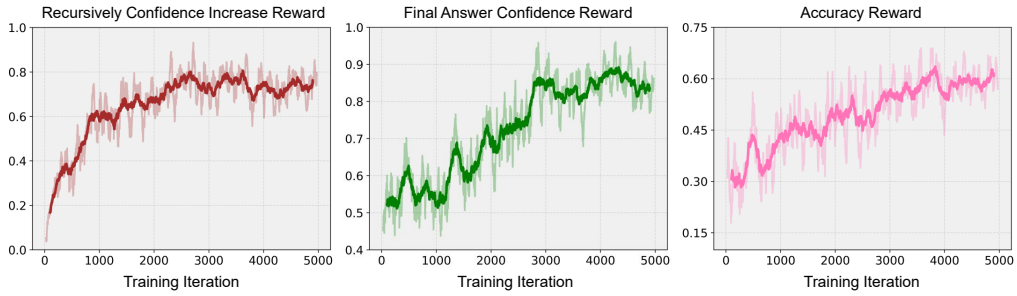


Figure 5. Training curves showing the progression of three reward signals—recursively confidence increase reward, last answer's confidence reward, and accuracy reward—over iterations during GRPO [70]. All rewards show consistent upward trends, indicating effective recursive refinement.

mediate reasoning steps, followed by extensions such as Program-of-Thoughts [18], Tree of Thoughts [90], and Graph of Thoughts [12] that explore larger reasoning spaces through structured search. While these methods improve robustness, they typically rely on external sampling or reranking rather than intrinsic self-evaluation.

Several reinforcement-learning-based methods introduce iterative refinement signals, including Reflexion [71] and Self-Consistency [80], but they operate outside the Think–Answer framework or depend on majority voting instead of introspective certainty. Recent Think-Answer models—DeepSeek-R1 [35] and OpenAI o1 [42]—demonstrate that separating "thinking" from "answering" can yield strong results in mathematical reasoning and programming. Similar trends are observed in multimodal reasoning, where models such as LMM-R1 [66], Vision-R1 [40], R1-V [16], and R1-Zero [102] adopt slow-thinking trajectories for visual problem solving.

However, almost all existing Think–Answer systems adopt a *single-pass* reasoning trajectory: after producing one Think–Answer pair, the model terminates inference even when its reasoning is uncertain or contains explicit self-reflective cues (e.g., "Oops!" and "Let me try again"). Sampling-based re-ranking strategies partially mitigate this issue but do not enable the model to internally assess confidence or decide whether further reasoning is needed.

Recursive and Confidence-Guided Refinement. Although iterative refinement has been explored through external verification [80] or heuristic feedback [71], prior works lack a principled mechanism to internally estimate the correctness of reasoning and dynamically invoke additional reasoning cycles. No existing Think–Answer model explicitly incorporates a confidence predictor to guide recursive inference-time reasoning.

In contrast, R-TAP introduces a confidence-aware recursive Think–Answer process that enables models to introspect, detect low-confidence reasoning, and selectively re-engage in additional reasoning cycles. By measuring both the recursive increase in confidence across reasoning cycles and the confidence of the final prediction, R-TAP offers a unified and efficient mechanism for self-corrective reasoning in both LLMs and VLMs, moving beyond the static, single-pass nature of prior approaches.

3. R-TAP: Recursive Think–Answer Process

R-TAP enables a model to perform confidence-guided recursive reasoning, allowing it to (i) evaluate the reliability of its reasoning, (ii) continue reasoning when uncertain, and (iii) terminate early when sufficiently confident. This section presents the overall framework, the Confidence Generator, the recursive generation scheme, the confidence-based reward design, and implementation details.

3.1. Problem Definition and Overall Framework

Given a question q, conventional single-step reasoning LLMs/VLMs produce exactly single Think-Answer pro-

Table 3. Performance comparison across various models including our proposed AZR variants on coding and mathematical reasoning benchmarks: HumanEval [17], MBPP [9], LiveCodeBench (LCB) [9], AIME [65], AMC [1], MATH500 [3], Minerva [4], and OlympiadBench [36].

Model	Base	HEval	MBPP	LCB	AME24	AME25	AMC	M500	Minerva	Olympiad	CAvg	MAvg	AVG
Base Models													
Qwen2.5-7B [87]	-	73.2	65.3	17.5	6.7	3.3	37.5	64.8	25.0	27.7	52.0	27.5	39.8
Qwen2.5-7B-Ins [87]	-	75.0	68.5	25.5	13.3	6.7	52.5	76.4	35.7	37.6	56.3	37.0	46.7
Qwen2.5-7B-Coder [87]	Coder	80.5	69.3	19.9	13.3	6.7	40.0	54.0	17.1	21.9	56.6	33.9	40.2
Qwen2.5-7B-Math [87]	-	61.0	57.9	16.2	10.0	16.7	42.5	64.2	15.4	28.0	45.0	29.5	37.3
Zero-Style Reasoners for Code													
AceCoder-RM [96]	Ins	79.9	71.4	23.6	20.0	6.7	50.0	76.4	34.6	36.7	58.3	37.4	47.9
AceCoder-Rule [96]	Ins	77.4	69.0	19.9	13.3	6.7	50.0	76.0	37.5	37.8	55.4	36.9	46.2
AceCoder-RM [96]	Coder	78.0	66.4	27.5	13.3	6.7	50.0	62.6	29.4	29.0	57.3	27.5	42.4
AceCoder-Rule [96]	Coder	80.5	70.4	29.0	6.7	6.7	37.5	62.8	27.6	27.4	60.0	28.5	44.8
CodeR1-LC2k [52]	Ins	81.7	71.7	28.1	13.3	3.3	40.0	74.0	35.6	36.7	60.3	35.8	48.6
CodeR1-12k [52]	Ins	81.1	73.5	29.9	13.3	3.3	37.5	74.0	35.7	36.9	61.3	33.5	47.4
PRIME-Zero [23]	Coder	49.4	51.1	11.0	23.3	**23.3**	**67.5**	**81.2**	37.9	41.8	37.2	45.8	41.5
AZR [101]	Coder	83.5	69.6	31.7	20.0	10.0	57.5	72.6	36.4	38.2	61.6	39.1	50.4
AZR-R-TAP	Coder	**89.3**	**75.9**	**38.2**	**27.4**	18.2	64.9	78.1	**44.3**	**46.6**	**67.8**	**46.6**	**57.2**
Zero-Style Reasoners for Math													
SimpleRL-Zoo [97]	Base	62.3	63.2	25.5	**20.0**	3.3	**67.5**	80.7	37.1	41.0	54.0	38.5	46.5
Oat-Zero [55]	Math	62.2	59.0	15.2	16.7	3.3	66.7	65.5	44.3	44.9	51.6	35.4	43.4
ORZ [38]	Base	80.5	64.3	22.0	13.3	16.7	60.0	81.8	32.7	**45.0**	55.6	**41.6**	48.6
AZR [101]	Base	71.3	69.1	25.3	13.3	13.3	52.5	74.4	38.2	38.2	55.2	39.0	46.8
AZR-R-TAP	Base	**78.5**	**76.7**	**32.4**	19.0	**20.1**	59.8	**81.2**	**45.7**	44.4	**62.5**	38.4	**50.5**

Table 4. Comparison of mathematical and coding benchmark performance across various instruction-tuned reinforcement-tuned and R-TAP integrated models. Benchmarks include AIME [65], AMC [1], MATH500 [3], MinervaMath [4], OlympiadBench [36], LeetCodeBench [22], and LiveCodeBench [43].

Method	AIME 2024	AMC	MATH500	MinervaMath	OlympiadBench	LeetCodeBench	LiveCodeBench	Avg.
GPT-4o [41]	9.3	45.8	76.4	36.8	43.3	**58.9**	**48.8**	45.6
Llama-3.1-70B-Inst. [32]	20.0	37.3	65.0	37.1	30.5	35.0	34.4	37.0
Qwen2.5-Math-7B-Inst. [87]	13.3	50.6	79.8	34.6	40.7	11.7	11.3	34.6
Eurus-2-7B-SFT [93]	3.3	30.1	66.2	32.7	29.8	21.7	17.8	28.8
RLOO [7]	20.0	47.0	73.2	36.4	35.4	28.3	26.7	36.9
Eurus-2-7B-PRIME [93]	20.0	50.6	78.2	39.3	40.3	31.1	27.5	41.0
Eurus-2-7B-PRIME-R-TAP	**28.3**	**57.5**	**83.5**	**43.8**	**47.4**	38.6	31.8	**47.2**

Table 5. GRPO-trained models on MathVerse [100], MathVision [78], MathVista [58], WeMath [67], and HallusionBench [50].

Model	MathVerse	MathVision	MathVista	WeMath	HallusionBench	Avg.
R1-VL-7B [98]	40.0	24.7	63.5	-	-	-
Vision-R1-7B [40]	52.4	-	73.5	-	-	-
R1-OneVision-7B [89]	46.1	22.5	63.9	62.1	65.6	52.0
OpenVLThinker-7B [24]	48.0	25.0	71.5	67.8	70.8	56.5
MM-Eureka-Qwen-7B [61]	50.5	28.3	71.5	65.5	68.3	56.8
ADORA-7B [33]	50.1	27.6	71.1	67.1	53.1	53.8
ThinkLite-7B-VL [81]	50.2	27.6	72.7	69.2	71.0	58.1
VLAA-Thinker-Qwen2.5-7B [14]	49.9	26.9	68.8	67.9	68.6	56.4
Qwen2.5-VL-7B-Instruct [10]	46.2	25.0	67.5	63.1	71.2	53.3
+ Vanilla GRPO [70]	50.7	28.5	71.7	68.6	69.8	57.9
+ NoisyRollout [54]	52.8	28.9	72.9	71.9	70.8	59.5
+ **R-TAP**	**60.1**	**35.3**	**79.4**	**78.2**	**77.9**	**66.2**

cess:

$$o^{(1)} \sim \pi_\theta(o \mid q), \tag{1}$$

and terminate immediately, even when the produced reasoning is uncertain or incorrect. Such models lack an internal mechanism for introspection or self-initiated refinement.

R-TAP generalizes this paradigm by allowing the model to recursively generate recursive Think-Answer responses $\mathcal{O}$ as follows:

$$\mathcal{O} = \{o^{(1)}, o^{(2)}, \ldots, o^{(T)}\} \sim \pi_\theta(\mathcal{O} \mid q), \quad \text{such that} \quad o^{(t+1)} \sim \pi_\theta(o \mid q, \{o^{(i)}\}_{i=1}^{t}), \tag{2}$$

where T denotes the recursion depths. During training, T is fixed to do efficient batch sampling, while at inference time the model internally determines whether to continue or terminate.

Similar to prior single-pass Think-Answer models [35], R-TAP also trains the model by maximizing the objective of GRPO [70] based on G generated samples, which can be written as follows:

$$\max_\theta \mathbb{E}_{q\sim\mathcal{D},\{\mathcal{O}_i\}_{i=1}^{G}\sim\pi_\theta(\mathcal{O}|q)}[\mathbb{E}_i[\min\left(r_i A_i, \text{clip}(r_i, 1-\epsilon, 1+\epsilon)A_i\right) - \beta\mathcal{D}_{\text{KL}}(\pi_\theta\|\pi_{\text{ref}})]], \tag{3}$$

where the policy ratio r_i and advantage A_i are defined by

$$r_i = \frac{\pi_\theta(\text{O}_i \mid q)}{\pi_{\theta_{\text{old}}}(\text{O}_i \mid q)}, \qquad A_i = \frac{R_i - \mathbb{E}[\{R_i\}_{i=1}^{G}]}{\sigma[\{R_i\}_{i=1}^{G}]}, \tag{4}$$

with R_i denotes the total reward for recursive trajectory $\mathcal{O}_i$ which will be explained in next section. This objective encourages the model to engage in further recursive reasoning when it internally detects low confidence, while allowing it to stop once its confidence reaches a satisfactory level.

3.2. Confidence Generator

A central component of R-TAP is the *Confidence Generator* $\mathbb{C}_\phi$, which provides the reliability of each response in recursive Think-Answer trajectory. Importantly, $\mathbb{C}_\phi$ is used only during R-TAP training and removed at inference, so R-TAP introduces no inference-time cost.

Architecture. Given a question q and one of Think-Answer responses: $o^{(t)}$ in Eq. (2), the Confidence Generator $\mathbb{C}_\phi$ outputs a scalar confidence score from zero to one continuous value:

$$\text{Conf}^{(t)} = \mathbb{C}_\phi(q, o^{(t)}). \tag{5}$$

We build Confidence Generator from π_{ref} structure in the beginning but replace the language head with a confidence head $\mathbb{R}^{d\times 1}$ (d: hidden dimension), followed by a sigmoid activation.

Supervised pre-training. Prior to R-TAP training, we train $\mathbb{C}_\phi$ to predict confidence of the given response. To implement it, we utilize the binary classification for whether the response is correct compared with ground-truth label. For each question q, the model generates N single Think-Answer process samples. Let $\{o_i^{\text{correct}}\}_{i=1}^{K}$ denote samples with correct answers and $\{o_i^{\text{wrong}}\}_{i=1}^{N-K}$ denote incorrect ones. We optimize $\mathbb{C}_\phi$ using binary classification objective:

$$\max_\phi[\frac{1}{K}\sum_{i=1}^{K}\log\mathbb{C}_\phi(q, o_i^{\text{correct}}) + \frac{1}{N-K}\sum_{i=1}^{N-K}\log\big(1-\mathbb{C}_\phi(q, o_i^{\text{wrong}})\big)], \tag{6}$$

where we technically set $K \approx N/2$ for balanced training. This step equips $\mathbb{C}_\phi$ with the ability to evaluate the reliability of recursive Think-Answer trajectories.

3.3. Confidence-Based Reward Design

R-TAP introduces two complementary confidence-driven rewards that (1) encourage refinement when necessary and (2) encourage termination when sufficiently confident.

Table 6. Performance of closed-source and open-source multimodal models on the R1-OneVision benchmark [89]. The benchmark is organized by diverse education level and subject category.

Model	Avg	Grade			Category					
		Junior High School	High School	College	Social Test	Math	Physics	Chemistry	Biology	Deduction
Closed-source										
GPT-4o [41]	49.6	51.3	56.2	45.3	26.5	41.3	52.5	71.4	63.4	26.5
Gemini-2.0-Flash [73]	59.1	56.0	65.9	61.2	39.8	52.3	64.4	74.3	67.2	39.8
Claude-3.5 [8]	52.1	56.0	55.9	49.4	30.6	46.5	54.3	66.7	65.7	30.6
Open-source										
MiniCPM-o-2.6 [39]	30.4	33.4	31.7	21.2	31.6	24.2	31.7	30.5	41.8	31.6
InternVL2.5-8B [19]	29.5	33.1	30.6	21.8	27.6	26.3	24.8	32.4	46.3	27.6
InternVL2.5-8B-MPO [19]	32.5	37.4	33.6	24.7	26.5	28.7	29.9	41.0	44.8	26.5
Qwen2-VL-7B [79]	30.0	35.4	32.5	25.9	26.5	26.3	28.1	30.5	45.5	26.5
Qwen2.5-VL-7B [79]	32.1	35.3	37.1	25.9	19.4	31.5	27.3	39.0	47.0	19.4
DeepSeek-VL2 [85]	29.8	34.4	30.9	18.8	30.6	23.5	28.4	29.5	47.8	30.6
R1-Onevision-7B [89]	36.2	40.1	39.5	27.6	26.5	33.0	30.2	49.5	53.0	26.5
R1-Onevision-7B-R-TAP	**42.4**	**47.4**	**45.0**	**34.9**	**31.4**	**39.4**	**36.9**	**55.9**	**59.6**	**31.2**
Qwen2.5-VL-72B [10]	52.0	54.3	56.7	54.1	23.5	48.9	55.8	63.8	63.4	23.5

Table 7. Comparing the performances on AIME2024 [65], AIME2025 [65], LiveCodeBench [43], and Arena-Head [75].

Model	AIME2024	AIME2025	LiveCodeBench (v5, 2024.10–2025.02)	Arena-Hard
AM-Thinking-v1 (Dense, 32B) [44]	85.3	74.4	70.3	92.5
AM-Thinking-v1-R-TAP (Dense, 32B)	**90.1**	**79.6**	**76.5**	**94.3**
Qwen3-235B-A22B (MoE, 235B) [86]	85.7	81.5	70.7	95.6
Qwen3-235B-A22B-R-TAP (MoE, 235B)	**90.9**	**88.7**	**77.6**	**97.1**
Qwen3-32B (Dense, 32B) [86]	81.4	72.9	65.7	93.8
Qwen3-32B-R-TAP (Dense, 32B)	**87.9**	**78.2**	**71.0**	**95.6**
DeepSeek-R1 (MoE, 671B) [35]	79.8	70.0	64.3	93.2
Nemetron-Ultra-253B (Dense, 256B) [11]	80.8	72.5	68.1	87.0
OpenAI-o1 (2024-12-17) [42]	74.3	79.2	63.9	92.1
OpenAI-o3-mini (Medium) [63]	79.6	74.8	66.3	89.0
Gemini2.5-Pro [73]	92.0	86.7	70.4	96.4

Table 8. Comparison of closed-source, general-purpose open-source, reasoning-focused, and proposed models on math-related benchmarks: MathVista [58], MathVerse [100], MathVision [78], OlympiadBench [36], WeMath [67].

Model	MathVista	MathVerse	MathVision	OlympiadBench	WeMath
Closed-Source Models					
Claude3.7-Sonnet [8]	66.8	52.0	41.3	48.9	72.6
GPT-4o [41]	63.8	50.2	30.4	35.0	68.8
o1 [42]	73.9	57.0	60.3	68.0	98.7
Gemini2-flash [73]	70.4	59.3	41.3	51.0	71.4
Open-Source General Models					
InternVL2.5-VL-8B [19]	64.4	39.5	19.7	12.3	53.5
Qwen-2.5-VL-7B [10]	68.2	47.9	25.4	20.2	62.1
InternVL2.5-VL-38B [19]	71.9	49.4	31.8	32.0	67.5
Qwen-2.5-VL-32B [10]	74.7/71.7	49.9	40.1	30.0	69.1
InternVL2.5-VL-78B [19]	72.3	51.7	32.2	31.1	66.3
Qwen-2.5-VL-72B [10]	74.8	57.6	38.1	40.4	72.4
Open-Source Reasoning Models					
InternVL2.5-8B-MPO [19]	68.9	35.5	21.5	7.8	53.5
InternVL2.5-38B-MPO [19]	73.8	46.5	32.3	25.6	66.2
QVQ-72B-Preview [73]	71.4	48.2	35.9	33.2	65.4
ADORA-7B [33]	73.5	50.1	23.0	20.1	64.2
R1-Onevision-7B [89]	64.1	47.1	29.9	17.3	61.8
OpenVLThinker-7B [24]	70.2	47.9	25.3	20.1	64.3
MM-Eureka-7B [61]	73.0	50.3	26.9	20.1	66.1
MM-Eureka-7B-R-TAP	**79.3**	**56.1**	**31.7**	**27.5**	**71.0**
MM-Eureka-32B [61]	74.8	56.5	34.4	35.9	73.4
MM-Eureka-32B-R-TAP	**80.2**	**61.8**	**39.9**	**41.2**	**79.3**

Recursive confidence increase reward. To reward meaningful refinement across recursive steps, we define:

$$R_{\text{Increase}} = \frac{1}{M-1} \sum_{t=1}^{M-1} \mathbb{1}\left[\text{Conf}^{(t+1)} > \text{Conf}^{(t)}\right], \quad (7)$$

where M is the effective recursion depth. For example, if the third Think-Answer response is correct but the previous ones are not, then $M = 3$ is satisfied, but if the first Think-Answer response is correct, because $M = 1$, we define this case to one reward value. Note that M cannot be larger than T such that $M \leq T$ because we pre-define T for effective batch sampling to generate recursive Think-Answer responses.

Final answer confidence reward. The final answer must be sufficiently confident:

$$R_{\text{Final}} = \mathbb{1}\left[\text{Conf}^{(M)} \geq \tau\right], \quad (8)$$

where τ is a preset threshold.

Combined reward. The total reward is simply calculated by the addition, which is written as follows

$$R = R_{\text{Increase}} + R_{\text{Final}} + R_{\text{Format}} + R_{\text{Answer}} + R_{\text{Length}}, \quad (9)$$

where we equally use the conventional rewards used in Open-R1 [28]. R_{Format} is the reward for Think-Answer format, R_{Answer} is for model response' correctness, and R_{Length} is for a soft length penalty. Appendix B describes the algorithms for R-TAP.

In summary, R-TAP provides broadly applicable mechanism for any LLMs and VLMs to enable recursive, confidence-guided reasoning within existing Think–Answer architectures. Furthermore, since the Confidence Generator is used only during training, R-TAP preserves the inference-time efficiency of standard single-pass models while equipping them with the ability to internally assess uncertainty and selectively refine their own reasoning for more stable yet faster. This makes R-TAP a scalable and practical solution for improving the stability, reliability, and depth of reasoning in modern models—bridging the gap between rigid single-pass inference and fully self-corrective reasoning systems.

4. Experiments

4.1. Implementation Details

We conduct training and evaluation of R-TAP primarily on NVIDIA A100 80GB GPUs. To ensure fast text generation during training, we employ vLLM [47], which is built on PagedAttention. In the pre-training step of Confidence Generator, vLLM [47] is used to generate $N = 128$ responses for each question for both LLMs and VLMs. We train the Confidence Generator under DeepSpeed engine with ZeRO-3 [68] with AdamW optimizer [57], applying a linearly decayed learning rate from 1e-5 to 1e-6 to pre-train the confidence generator. Next step is training LLMs/VLMs as well as the Confidence Generator, where we also leverage vLLM [47] for online text generation and use DeepSpeed with ZeRO-3 to update both models with a fixed learning rate of 1e-6. During this phase, we use $T = 4$ recursion depths, for which we generate $G = 12$ response outputs. From this setting, we update target models for 12 GRPO [70] iterations per each training iteration by using $\epsilon = 0.2$ and $\beta = 0.04$ in Eq. (3). For diverse responses, we set the generation hyperparameters to temperature=1.0, top-p=0.95, top-k=50, and repetition penalty=1.05. In addition, we set τ to 0.55 and we find its optimal τ by doing greedy

Table 9. Performance of Llama3.1 [32], DeepSeek-Math [70], Mistral [45], and Qwen-2.5 [87] models before and after SimpleRL-Zoo [97] fine-tuning and R-TAP integration across various mathematical reasoning benchmarks. Benchmarks include GSM8K [20], MATH500 [3], Minerva [4], OlympiadBench [36], AIME [65], and AMC [1]. Note that, the selection of evaluation benchmarks and their performance results, and the selection of the compared models are referred to the manuscript of SimpleRL-Zoo [97].

Model	GSM8K	MATH500	Minerva Math	Olympiad Bench	AIME24 (P@1)	AIME24 (Avg@32)	AMC23
Llama, DeepSeek and Mistral Models							
Mistral-v0.1-7B [45]	21.2	4.2	4.0	2.4	0.0	0.0	0.0
+ SimpleRL-Zoo	75.0	15.8	6.6	4.1	0.0	0.2	10.0
+ R-TAP	**81.2**	**21.3**	**13.9**	**10.0**	**6.4**	**9.5**	**17.4**
Llama-3.1-8B [27]	39.7	13.6	4.8	3.1	0.0	0.2	2.5
+ SimpleRL-Zoo	79.2	23.0	9.6	5.3	0.0	0.2	15.0
+ R-TAP	**86.5**	**30.4**	**16.3**	**13.7**	**6.7**	**7.9**	**22.3**
DeepSeek-Math-7B [70]	28.4	19.4	5.5	4.7	0.0	0.0	10.0
+ SimpleRL-Zoo	78.5	39.6	21.0	12.6	3.3	0.6	20.0
+ R-TAP	**83.7**	**45.2**	**28.4**	**20.1**	**10.2**	**8.2**	**27.5**
Mistral-Small-24B [45]	78.6	43.6	10.7	11.6	3.3	0.5	17.5
+ SimpleRL-Zoo	92.0	70.6	36.8	36.6	16.7	13.1	45.0
+ R-TAP	**95.4**	**77.7**	**40.2**	**43.2**	**23.4**	**18.5**	**51.1**
Qwen Series Models							
Qwen-2.5-0.5B [87]	36.7	15.8	4.8	2.8	0.0	0.3	12.5
+ SimpleRL-Zoo	49.5	34.4	10.3	8.9	0.0	0.7	22.5
+ R-TAP	**54.3**	**39.8**	**16.6**	**14.2**	**6.4**	**6.9**	**28.3**
Qwen-2.5-1.5B [87]	55.7	29.6	6.6	6.5	0.0	0.1	12.5
+ SimpleRL-Zoo	74.4	59.0	20.2	21.0	6.7	4.2	35.0
+ R-TAP	**79.2**	**64.4**	**27.1**	**28.2**	**13.0**	**11.5**	**41.7**
Qwen-2.5-7B [87]	88.2	64.6	25.7	30.1	3.3	0.3	30.0
+ SimpleRL-Zoo	91.7	78.2	38.6	40.4	20.0	15.6	62.5
+ R-TAP	**96.0**	**83.5**	**44.8**	**46.9**	**28.1**	**22.3**	**69.4**
Qwen-2.5-Math-7B [88]	86.5	63.6	12.5	25.8	13.3	8.6	42.5
+ SimpleRL-Zoo	90.2	80.2	37.5	39.0	40.0	24.0	70.0
+ R-TAP	**94.2**	**86.7**	**42.3**	**46.1**	**45.6**	**29.3**	**78.1**
Qwen-2.5-14B [87]	91.6	65.4	24.3	33.5	6.7	3.4	37.5
+ SimpleRL-Zoo	94.4	80.2	40.4	44.9	23.3	14.2	57.6
+ R-TAP	**97.8**	**86.3**	**48.1**	**52.3**	**30.5**	**19.8**	**61.7**
Qwen-2.5-32B [87]	92.9	68.6	27.9	31.1	10.0	4.5	45.0
+ SimpleRL-Zoo	95.9	82.4	42.6	46.4	36.7	27.2	67.5
+ R-TAP	**97.9**	**87.9**	**47.3**	**52.3**	**42.9**	**32.0**	**71.8**

search from zero to one value by 0.05 points. For evaluation, we remove the confidence generator and use only the trained LLMs/VLMs. We keep the default generation hyperparameters of backbone LLMs/VLMs.

4.2. Validating R-TAP

For LLMs, we train all the parameters, but for VLMs, we do not train vision encoder parts due to some observations of performance degradation. For selecting training dataset, we employ same training dataset on which LLMs/VLMs is trained for think-answer process, except some LLMs that do not release their own training dataset such as Phi-4-reasoning series [5], MiMo [72], Qwen3 [86], and AM-Thinking-v1 [44]. For these models, we instead gather Open-R1-Math (220K) and codeforce-cot (100K) in Open-R1 [28], and PRIME (481K) [23] covering math and code as well, and train them by using algorithm of R-TAP.

As shown in Fig. 1, R-TAP-applied Phi-4-reasoning models have shown dramatic improvements, thereby closing to OpenAI-o1 and -o3 models. Figure 5 represents the reward graph of Phi-4-reasoning-plus [5] during training R-TAP, indicating stable training. To see more generalizability and applicability, we employ several LLMs and their dataset. First, we consider R1-Distill-Qwen-1.5B and -7B [35] and Oat-Zero-1.5B, -7B [56], AZR-Base-7B and AZR-Coder-7B [101] as backbone models and we train them by R-TAP on Open-Reasoner-Zero-57K [37]. In addition, we apply R-TAP to SimpleRL-Zoo [97] in Tab. 9 and PRIME [23] by using each their own curated data (57K and 481K, respectively). Remarkably, Tab. 1-Tab. 4 shows dramatic improvements of LLMs with R-TAP as well. We additionally employ recently released LLMs: Qwen3 [86] and AM-Thinking-v1 [44], and we observe their consistent dramatic performance improvements by R-TAP in Tab. 7, compared with state-of-the-art open- and closed-source reasoning LLMs.

To extend its effectiveness to VLMs, we apply R-TAP to Skywork-R1V2 [84] by Skywork-OR1-RL-120K, Geometry-3K, R1-OneVision-155K, and MMK12-16K. Notably, Fig. 2 shows its consistent performance improvements despite multimodality. Besides, we employ Noisy-Rollout-7B [54], R1-OneVision-7B [89], and MM-Eureka-7B and -32B [61] and train them on their own training dataset: Geometry-3K, R1-OneVision-155K, and MMK12-16K. Tab. 5, Tab. 6, and Tab. 8 shows their performance improvements as well.

4.3. Ablation Studies

We conduct comprehensive ablation studies to validate the effect of each R-TAP component in Tab. 10 and Tab. 11, superiority of confidence estimation method in Tab. 12 and Tab. 13, and the effect of recursion depths in Tab. 14 and Tab. 15. In this study, we deal with Phi-4-reasoning-plus [5] for LLM and Skywork-R1V2 [84] for VLM. We analyze the effect of removing confidence increase reward R_{Increase}, final answer confidence reward R_{Final} and answer reward R_{Answer}. Tab. 10 and Tab. 11 show that all components of R-TAP are essential for achieving highest accuracy in both language-only and multi-modal settings.

We further show the superiority of our Confidence Generator design by comparing with four recent confidence estimation approaches: (a) Calibration via ECE [31, 34] (b) Se-

Table 10. Effect of each R-TAP component on LLM

Method	$\mathbb{C}_\phi$	R_{Increase}	R_{Final}	R_{Answer}	AIME25	HMMT Feb25	OmniMath	GPQA	LiveCodeBench	Avg
Baseline	✗	✗	✗	✓	78.0	53.6	81.9	69.3	65.9	69.7
R-TAP	✓	✓	✗	✓	80.1	56.2	84.0	71.0	70.1	72.3
R-TAP	✓	✗	✓	✓	81.5	57.8	85.1	73.2	72.0	73.9
R-TAP	✓	✓	✓	✓	**83.7**	**60.3**	**86.2**	**76.7**	**72.1**	**75.8**
R-TAP	✓	✓	✓	✗	61.2	44.7	59.9	53.0	51.8	56.1

Table 11. Effect of each R-TAP component on VLM

Method	$\mathbb{C}_\phi$	R_{Increase}	R_{Final}	R_{Answer}	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg
Baseline	✗	✗	✗	✓	73.6	74.0	62.6	49.0	52.0	62.2
R-TAP	✓	✓	✗	✓	76.1	78.0	66.4	53.0	54.4	65.4
R-TAP	✓	✗	✓	✓	77.8	80.2	68.3	55.3	58.1	67.9
R-TAP	✓	✓	✓	✓	**78.2**	**82.3**	**69.4**	**56.8**	**59.2**	**69.2**
R-TAP	✓	✓	✓	✗	53.2	50.7	45.9	41.3	52.4	48.7

Table 12. Confidence Estimation Methods for LLM

LLMs	AIME25	HMMT Feb25	OmniMath	GPQA	LCB	Avg
Phi-4-reasoning-plus	78.0	53.6	81.9	69.3	65.9	69.7
Phi-4-reasoning-plus (a)	81.5	57.0	85.7	73.0	69.3	73.3
Phi-4-reasoning-plus (b)	83.0	55.8	86.5	73.5	68.7	73.5
Phi-4-reasoning-plus (c)	82.5	57.0	87.2	72.8	70.5	74.0
Phi-4-reasoning-plus (d)	**84.2**	57.3	**87.5**	75.1	70.4	74.9
Phi-4-reasoning-plus-R-TAP	83.7	**60.3**	86.2	**76.7**	**72.1**	**75.8**

Table 13. Confidence estimation methods for VLM

VLMs	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg
R1V2-38B	73.6	74.0	62.6	49.0	52.0	62.2
R1V2-38B (a)	75.1	78.0	65.8	53.5	57.6	66.0
R1V2-38B (b)	76.3	77.5	67.2	53.0	58.5	66.5
R1V2-38B (c)	77.0	79.0	68.5	54.2	57.3	67.2
R1V2-38B (d)	**79.5**	80.3	**70.8**	**58.2**	55.7	68.9
R1V2-38B-R-TAP	78.2	**82.3**	69.4	56.8	**59.2**	**69.2**

Table 14. Recursion depths for LLM

LLMs	T	AIME25	HMMT Feb25	OmniMath	GPQA	LiveCodeBench	Avg
Phi-4-reasoning-plus	1	78.0	53.6	81.9	69.3	65.9	69.7
Phi-4-reasoning-plus	2	78.7	54.2	82.4	70.4	67.3	71.0
Phi-4-reasoning-plus	3	79.2	55.0	82.8	71.0	67.9	71.2
Phi-4-reasoning-plus	4	80.3	56.4	83.6	72.5	69.1	72.5
Phi-4-reasoning-plus-R-TAP	2	80.9	57.1	84.1	73.3	68.6	72.6
Phi-4-reasoning-plus-R-TAP	3	82.3	58.9	85.0	75.1	70.3	74.9
Phi-4-reasoning-plus-R-TAP	4	**83.7**	**60.3**	**86.2**	**76.7**	**72.1**	**75.8**

Table 15. Recursion depths for VLM

VLMs	T	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg
R1V2-38B	1	73.6	74.0	62.6	49.0	52.0	62.2
R1V2-38B	2	74.3	75.1	63.3	50.4	53.1	63.0
R1V2-38B	3	74.8	75.7	64.0	51.1	53.9	63.4
R1V2-38B	4	75.6	76.6	64.9	52.2	54.6	64.0
R1V2-38B-R-TAP	2	76.1	77.0	65.3	52.7	55.0	64.1
R1V2-38B-R-TAP	3	77.5	79.4	67.2	54.9	57.8	67.4
R1V2-38B-R-TAP	4	**78.2**	**82.3**	**69.4**	**56.8**	**59.2**	**69.2**

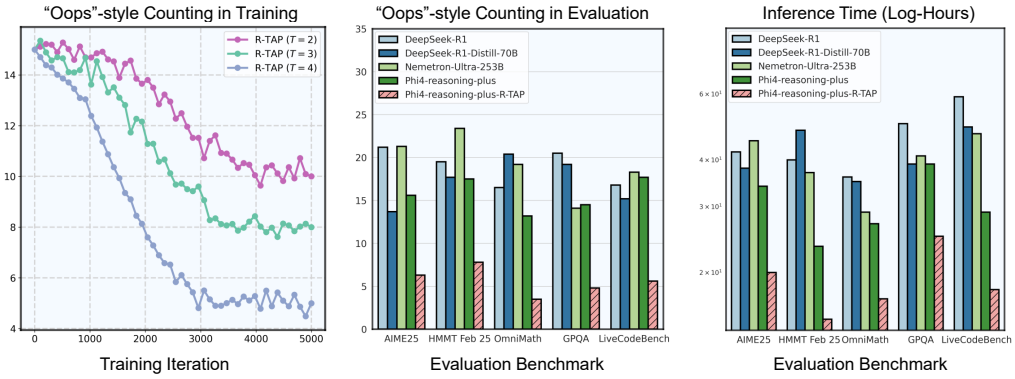


Figure 6. Impact of R-TAP on reducing the number of "Oops"-style words – which corresponds to the number of erroneous reasoning – and its effect on substantially reducing inference time. (Left) Negative correlation between the number of erroneous reasoning and R-TAP train iterations. (Center) Evaluation result on the significant reduction of erroneous reasoning by applying R-TAP. (Right) Evaluation result on the substantial reduction of inference time due to the reduction of erroneous reasoning by applying R-TAP.

mantic Equivalence Entropy [46] (c) "IDK" token [21] and (d) Iterative LLM generation [48, 99]. Tab. 12 and Tab. 13 show that applying our method achieves highest accuracy. Tab. 14 and Tab. 15 suggest that increasing recursion depths of Think-Answer from 1 to 4 times makes higher accuracy.

Finally, we additionally analyze how R-TAP affects the reduction of erroneous reasoning behaviors and inference efficiency, as summarized in Fig. 6. The left plot shows that the number of "Oops"-style tokens—our proxy for erroneous reasoning—monotonically decreases as training progresses, and deeper Think–Answer recursion (larger T) leads to faster and greater reduction. The center plot demonstrates that this trend generalizes to evaluation benchmarks: R-TAP consistently lowers erroneous reasoning across all datasets compared to baseline and the other state-of-the-art models. In addition, the right plot reveals that this reduction in reasoning errors directly translates to substantial improvements in inference efficiency, yielding significantly shorter inference time (log-hours) across all benchmarks. These results confirm that R-TAP not only improves accuracy but also makes reasoning more stable and computationally efficient. Besides, we describe detailed comparison between performance and computation complexity about decoding token count and training time in Appendix C.

5. Discussion and Conclusion

We introduced **R-TAP**, a method that augments LLMs and VLMs with a learned confidence generator and two rewards: recursive confidence increase and final-answer confidence reward. These components enable models to iteratively refine their think–answer trajectories, yielding substantial performance gains and narrowing the gap between smaller models and much larger ones. R-TAP supports up to T recursive steps. However, due to the batch-dependent nature of current deep-learning implementations, all T trajectories must be generated in parallel even when a confident answer emerges early. This simplifies parallelization but introduces significant computational and memory overhead. We hope that R-TAP encourages further research on confidence-aware iterative reasoning and supports the development of more efficient, trustworthy, and general-purpose reasoning systems for both LLMs and VLMs.

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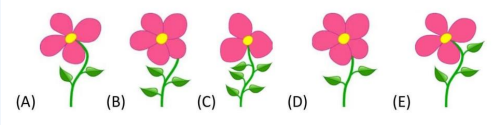
A. Full Responses for Multimodal Example

SYSTEM: You are a helpful AI Assistant that provides well-reasoned and detailed responses. You first think about the reasoning process as an internal monologue and then provide the user with the answer. Respond in the following format: <think> </think>\n<answer> </answer>

USER: The ladybird would like to sit on his flower. The flower has five petals and the stem has three leaves. On which flower should the ladybird sit?



ASSISTANT: <think> Let me inspect each flower carefully. Flower A appears to have exactly five petals, but its stem only sprouts two leaves. Flower B has a nice regular star of five petals—and I thought I spotted three little leaves tucked along its stem too. Flower C looks squashed, maybe only four broad petals and actually closer to four leaves as well. D has a perfect pentagon of petals but now that I look I remember only two leaves there. E seemed promising at first glance but I can't quite recall its leaf count. On balance, my initial gut is that D stands out with a clear five-petal bloom and I vaguely thought three leaves peeked out—but I must have misremembered. Still, I'll go with D for now. </think>
<answer>D</answer>

<think> **Re-checking D**, I see exactly two leaves—one on each side of the stem—so D definitely fails the "three leaves" criterion. The next candidate is E: I recall E had a single top-right leaf, another mid-left leaf, and one lower-left leaf, giving three in total, and its petals looked evenly spaced to make five. So E fits "five petals, three leaves," so my second guess will be E. </think>
<answer>E</answer>

<think> **Oops**—I've miscounted again. On closer inspection, E actually only has four fully visible petals because one is hidden behind the stem's curve. Meanwhile, B clearly displays five distinct petals and exactly three leaves along its stem (one low-left, one mid-right, one upper-left). Therefore the only flower meeting both criteria is B. </think>
<answer>B</answer>

Figure 7. Qualitative multimodal example of recursive think–answer process. The model refines its answer over multiple reasoning cycles by correcting initial counting errors and verifying object-level visual details, ultimately identifying the only flower that meets both criteria.

B. Algorithm of Recursive Think-Answer Process

Algorithm 1 Recursive Think-Answer Process for LLMs and VLMs

Require: Pre-trained Confidence Generator $\mathcal{C}_\phi$ and Pre-trained LLMs/VLMs $\pi_{\theta_{\text{init}}}$
Set reference model $\pi_{\text{ref}} \leftarrow \pi_{\theta_{\text{init}}}$
Set the training model $\pi_\theta \leftarrow \pi_{\theta_{\text{init}}}$
for sample a batch $\mathcal{B}$ in Dataset **do**
  Copy and freeze model $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta$
  Sample G outputs $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot|q)$, until M maximum recursive.
  Extract the responses G' in G outputs until the early correct responses.
  Replace the responses: $G \leftarrow G'$
  Compute rewards and advantages for G outputs by confidence generator & answer parsing
  for Confidence generator Updating iteration $= 1, 2, \cdots, \mu$ **do**
    Update $\mathcal{C}_\phi$ by using Eq. (6)
  end for
  for LLMs/VLMs Updating iteration $= 1, 2, \cdots, \mu$ **do**
    Update π_θ by using Equation Eq. (3)
  end for
end for

C. Detailed Comparison between Performance and Computation Complexity

Comparison with Related Refinement and Verification Methods. We compare R-TAP against four closely related baselines: (a) Reflexion [71], (b) Self-Consistency [80], (c) Self-Refine [59], and (d) Self-Verification [53]. All methods are evaluated under the same output token budget. Unlike iterative refinement or verification-based approaches, which rely on explicit self-feedback loops, majority voting, or external verification during inference, R-TAP learns when to continue or terminate reasoning via reinforcement learning. As a result, it removes the need for repeated trial-and-error generation at deployment. Although the confidence generator in R-TAP is trained with binary supervision, it does not function as a hard 0/1 classifier at inference. Instead, it produces a continuous signal in $[0, 1]$, which enables the model to measure confidence improvement across recursive steps and regulate reasoning depth smoothly.

	MaxTokenLimit	AIME25	HMMT Feb25	OmniMath	GPQA	LCB	Avg	Oops-style count	Decoding token count	Training Time
Phi-4-reasoning-plus	32768	78.0	53.6	81.9	69.3	65.9	69.7	15.7	14509.7	-
w/ Reflexion [71]	32768	80.2	56.1	83.4	71.3	66.5	71.5	18.9	15230.4	51 hours
w/ Self-Consistency [80]	32768	81.1	55.8	84.0	72.2	66.4	71.9	16.8	16012.9	58 hours
w/ Self-Refine [59]	32768	78.5	54.9	83.1	71.0	68.5	71.2	14.3	13890.6	45 hours
w/ Self-Verification [53]	32768	82.4	57.6	85.7	73.5	64.8	72.8	17.5	14890.3	51 hours
R-TAP	32768	**83.7**	**60.3**	**86.2**	**76.7**	**72.1**	**75.8**	**5.6**	**4378.8**	**28 hours**

	MaxTokenLimit	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg	Oops-style count	Output token count	Training Time
R1V2-38B	12000	73.6	74.0	62.6	49.0	52.0	62.2	17.2	9890.7	-
w/ Reflexion [71]	12000	74.5	74.8	63.0	49.5	53.7	63.1	17.5	10989.3	57 hours
w/ Self-Consistency [80]	12000	75.0	76.0	64.0	50.0	52.5	63.5	18.6	10678.5	60 hours
w/ Self-Refine [59]	12000	73.8	74.2	62.0	49.0	55.5	62.9	15.3	9938.2	59 hours
w/ Self-Verification [53]	12000	76.5	77.0	65.0	51.0	53.0	64.5	17.8	10183.8	58 hours
R-TAP	12000	**78.2**	**82.3**	**69.4**	**56.8**	**59.2**	**69.2**	**8.5**	**5789.4**	**39 hours**

Token Efficiency and Computational Cost. R-TAP does not increase total output tokens. On the contrary, it substantially reduces them. While recursive sampling is introduced during training, the learned policy discourages unnecessary self-corrections at inference time. As shown in this table, R-TAP reduces output tokens by approximately 2–3$\times$ compared to self-consistency, self-refine, and verification-based baselines under the same output token budget, while achieving higher accuracy.

T	G	AIME25	HMMT Feb25	OmniMath	GPQA	LiveCodeBench	Avg	Output token count	Training Time
1	4	76.1	52.1	80.4	67.9	64.4	68.2	15034.9	51 hours
1	8	77.2	52.9	81.2	68.6	65.2	69.0	15210.7	52 hours
1	12	78.0	53.6	81.9	69.3	65.9	69.7	14872.5	58 hours
2	12	80.9	57.1	84.1	73.3	68.6	72.6	9258.9	42 hours
3	12	82.3	58.9	85.0	75.1	70.3	74.9	6127.4	34 hours
4	12	83.7	60.3	86.2	76.7	72.1	75.8	4378.8	28 hours
4	8	83.0	59.7	85.7	76.0	71.4	75.2	4442.3	25 hours
4	4	82.2	58.9	85.0	75.2	70.6	74.4	4321.6	22 hours

T	G	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg	Decoding token count	Training Time
1	4	72.4	72.6	61.2	47.8	50.6	60.9	10342.7	55 hours
1	8	73.1	73.4	62.0	48.5	51.4	61.7	9927.3	62 hours
1	12	73.6	74.0	62.6	49.0	52.0	62.2	10168.9	68 hours
2	12	76.1	77.0	65.3	52.7	55.0	64.1	8234.6	55 hours
3	12	77.5	79.4	67.2	54.9	57.8	67.4	6912.1	46 hours
4	12	78.2	82.3	69.4	56.8	59.2	69.2	5789.4	39 hours
4	8	77.8	81.7	68.8	56.4	58.7	68.7	5698.2	34 hours
4	4	77.1	80.8	67.9	55.6	57.8	67.8	5861.5	30 hours

Effect of Majority Voting. We further evaluate self-consistency with varying voting numbers (N), under identical output token budgets (32768 for Phi-4-reasoning-plus and 12000 for R1V2-38B). As expected, increasing N consistently improves performance across all models. However, R-TAP-trained models already achieve strong performance with $N = 1$, and additional voting yields only marginal gains. This indicates that R-TAP does not replace self-consistency; rather, it learns a more stable single-sample reasoning policy, thereby reducing reliance on majority voting during inference.

Voting N	AIME25	HMMT Feb25	OmniMath	GPQA	LiveCodeBench	Avg
1	83.7	60.3	86.2	76.7	72.1	75.8
3	83.9	60.5	86.4	76.9	72.3	76.0
5	84.0	60.6	86.5	77.0	72.4	76.1
7	84.1	60.7	86.6	77.1	72.5	76.2
9	84.1	60.7	86.6	77.1	72.5	76.2
12	84.2	60.8	86.7	77.2	72.6	76.3
15	84.2	60.8	86.7	77.2	72.6	76.3
18	84.2	60.8	86.7	77.2	72.6	76.3

Voting N	MMMU	MathVista	OlympiadBench	MathVision	MMMU-Pro	Avg
1	78.2	82.3	69.4	56.8	59.2	69.2
3	78.3	82.4	69.5	56.9	59.3	69.3
5	78.3	82.4	69.5	56.9	59.3	69.3
7	78.4	82.5	69.6	57	59.4	69.4
9	78.5	82.6	69.7	57.1	59.5	69.5
12	78.5	82.6	69.7	57.1	59.5	69.5
15	78.5	82.6	69.7	57.1	59.5	69.5
18	78.5	82.6	69.7	57.1	59.5	69.5

Recap. R-TAP consists of four key components: (a) GRPO-based reinforcement learning for Think-Answer trajectories, (b) a recursive reward mechanism derived from both the confidence generator and intermediate results, (c) suppression of unnecessary Oops-style refinement, and (d) improved inference efficiency resulting from reduced refinement steps.

Importantly, while prior uncertainty- or refinement-based approaches use confidence signals for reranking, filtering, or verification after generation, R-TAP integrates confidence as an internal reinforcement signal during training. This signal continuously modulates recursive reasoning depth, directly shaping the learned reasoning policy instead of performing post-hoc correction at inference time.